Bebop as a Markov Chain

Speaker notes, slide by slide

Michele "Mike" Rubini · MSc Decision Science · Università di Bari
Dynamical Models for Social Behavior · Profs. Gonnella & Suma

How to use this

- The body text is what to **say**, written to be spoken rather than read aloud word for word. Say it in your own words; keep the numbers exact.
- *Italic crimson* lines are stage directions, not speech: what to point at, when to click.
- The boxed **IF ASKED** notes are for questions. Do not say them unprompted.
- The timings sum to **18 min 45 s**, leaving slack in a 20-minute slot. If you are running long, slides 4, 8 and 9 are the ones to compress to a single sentence each: they are illustrations, not results.
- After "Thank you" comes a Backup divider and three backup slides, in this order: **distribution evolution, the spectrum, coming back round**. Do not present them; go there only if a question sends you there.

#	SLIDE	TIME	CUMULATIVE
–	Title and research question	0:40	0:40
1	The corpus	0:55	1:35
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#	SLIDE	TIME	CUMULATIVE
19	Thank you	0:15	18:45
	Total	**18:45**	

Title Bebop as a Markov Chain 0:40

Good morning. My project is called *Bebop as a Markov Chain*.

The starting point is this. Jazz improvisation sounds free, but musicians do not describe it as free. They describe it as a language with a grammar: certain notes want to move to certain other notes, tension wants to resolve, and where you are in the bar changes what you are allowed to play. Those rules are taught in words, in every jazz school.

My question is whether the same rules can be recovered as **numbers**, from nothing but the notes, by treating a solo as a random walk on a graph.

The data is fifty Charlie Parker improvisations. No music theory is given to the model at any point. It only counts which state follows which. Everything that comes out has to come out of the counting.

1 The corpus 0:55

Fifty transcriptions from the Charlie Parker Omnibook, in MusicXML.

Each file is one tune: the head, meaning the written melody, followed by one or more improvised choruses. What makes the corpus usable is that the files also carry the chord symbols, so at every instant I know which chord is sounding underneath the note.

Extraction is three rules. Walk the part note by note. Pair each note with the most recent chord. Collapse tied notes, so a note held across a bar line is counted once, at its onset.

Rests are kept as events, because silence is a choice a soloist makes, but a run of consecutive rests collapses into one event, otherwise a two-bar silence would flood the model with rest-to-rest transitions.

That gives **23,143 events**: 21,590 notes and 1,553 rests.

Point at the table. One number here matters for the rest of the talk: **56 percent** of the corpus sits over dominant seventh chords. Bebop lives on the dominant, and that is going to show up in every result.

Point at the bottom-left panel. And the events are split almost evenly across the bar, with **55 percent falling offbeat**. The median solo is 426 events.

2 State space

This is the only modelling decision that really matters. Everything downstream is mechanical once the state is fixed.

A state is three coordinates.

First, scale degree. Not the note name, but the distance in semitones from the chord root up to the note. So a C played over a C7 is degree 0, and the same C over an F7 is degree 7. Rests get a sentinel, `R`.

Second, chord quality, in five classes: major seventh, minor seventh, dominant seventh, half-diminished and diminished.

Third, beat position. All fifty tunes are in four-four. An event either lands on one of the four beats, or it falls between them, and if it falls between them I label it by the beat it follows. So the "and of four", just before the bar line, is a different state from the "and of two".

Move to the right-hand column. Why these three.

Scale degree buys transposition invariance. The same phrase played over F7 and over B-flat 7 becomes the same sequence of states. Fifty solos in a dozen keys pool into one estimate instead of fragmenting into a dozen sparse ones.

Chord quality is needed because the same degree means different things. A flat seven over a dominant chord is the unstable note, the one that wants to fall. The same flat seven over a minor seventh chord is a chord tone, and it is stable. Same distance, opposite behaviour.

And beat position, because in bebop the beat carries the syntax. Chord tones land on the beat, tension goes between the beats. Without this coordinate that distinction is invisible to the model, and it turns out to be the single most informative coordinate of the three.

> IF ASKED
>
> **Why 12 semitones and not diatonic degrees?** Bebop is chromatic. Roughly half of the corpus is non-diatonic passing material, so a seven-degree encoding would throw away exactly what I am trying to model.
>
> **Why split the offbeat by which beat it follows?** Because "and of four" is functionally different from the others: it is the pickup into the next bar and usually into the next chord. Merging all offbeats into one class hides the strongest cell in the whole matrix, which is on slide 14.

3 The encoding, on a real solo 1:10

Here is the encoding on a real solo, the opening of *Anthropology*.

Walk left to right along the figure. The pickup bar starts with a rest over F7, so that is `(R, dom7, b1)`. Then Parker plays an F over F7, which is the root, offbeat on the "and of four": `(0, dom7, &4)`. The chord changes to B-flat major seventh, and he plays a B-flat, which is again degree 0 but now over a different quality, on beat one.

Look at the last three events on the right, because they are the whole thesis in miniature. He plays a **D-flat offbeat**, which encodes as `(3, maj7, &3)`, a minor third over a major chord. That note is outside the chord. It clashes. Then on beat four he plays a **D natural**, `(4, maj7, b4)`, the major third, which is a chord tone.

So: tension placed between the beats, resolution a semitone away, landing on the beat. That is a rule from a textbook, and here it is as two adjacent states. The rest of the talk is about whether the model finds that pattern on its own, across 23,143 events, without being told.

Point at the result line. The state space has at most 520 cells. **428** are actually visited, so about 54 visits per state on average. That recurrence is what makes the transition probabilities estimable at all.

And note what is deliberately **not** in the state: no duration, no octave, no dynamics. A state is a pitch class against a chord, on a beat, and nothing else.

> IF ASKED
>
> **Is there a recording?** Yes, the video on this slide is the take these bars come from.

4 What Parker never plays 0:30

Briefly, the 92 states that never occur, because they have two very different causes.

Ninety of them are just scarcity. Each chord quality has its own grid of 13 degrees by 8 beat positions, so 104 cells. Dominant seventh fills all 104 of its cells, because it carries more than half the corpus. Half-diminished and diminished carry about 165 events each, and 165 events cannot cover 104 cells. Those are empty for want of data, not for musical reasons.

The rest are structural. Some states are empty because the note simply clashes with the chord, for instance a major third over a minor chord. Parker does not play those, and no amount of extra data would fill them in.

I mention it because the distinction matters when reading the matrix on the next slide: an empty cell can mean "never happens" or "never observed", and those are not the same claim.

5 The operator, concretely

This is the object the whole project is about, and I want to be concrete about what it is before doing anything with it.

P-hat is a table with 428 rows and 428 columns. **Every row asks one question: Parker is on this state, where does he go next?**

Building it is pure counting. Walk the 23,143 events in order. Every time state i is followed by state j, add one to cell i, j. At the end, divide each row by its own total. That is the maximum likelihood estimate, and it is all there is to it.

The hat matters. **P-hat is an estimate.** It is what fifty solos say about a true operator I cannot observe. Every number I quote today is a sample statistic, not a law.

Two things to read off the picture. *Point at the full matrix.*

First, probabilities live **inside a row**. Each row sums to one across all 428 columns. So the numbers you see in the block on the right are all conditional: given that I am here, this is the chance I go there.

Second, the matrix is **97 percent empty, and most of that is metre, not music**. From the "and of four" the walk goes to beat one 69 percent of the time, or stays where it is 21 percent of the time, and it reaches beat three only 0.4 percent of the time. Most cells are empty because the beat cannot get there. I will come back to this at the end, because it is the main caveat on everything else.

Only if someone looks puzzled at the pink. On colour: dark means a move Parker actually plays, near-white means the model is confident the move does not happen, and pink means the model has almost no data for that row and is spreading probability evenly. Pink is ignorance, not structure.

> IF ASKED
>
> **Is P symmetric?** No, and it could not be. Symmetric would mean the flat seven falls to the third exactly as often as the third rises to the flat seven. Voice leading is directional, and the asymmetry is the result.
>
> **What about zero rows?** I add Laplace smoothing with alpha = 0.01. Without it some rows are all zeros, and then there is no unique stationary distribution and the hitting-time system is singular. The cost is that thin rows get pulled toward uniform, which is what the pink bands are.
>
> **How many transitions?** 21,540. Fifty tunes means fifty boundaries, and I do not count a transition across a tune boundary.

6 The Markov property 1:20

Before using the operator I should check the assumption that licenses it, and I want to be honest about the answer.

The Markov property says the next state depends only on where the line **is**, not on how it got there. Everything before the current event is discarded. If that is false, then Parker cannot be summarised as a table of "from, to", and the table on the previous slide is not entitled to exist.

So I tested it directly. *Point at the table.* Take one busy state: the fifth of a dominant chord, on the "and of four". It occurs 346 times with a known predecessor. Now split those 346 continuations by **what came before**.

Arriving from the sixth, the most likely next note is the root on the downbeat, 22 percent of the time.

Arriving from the third, the most likely next note is a different one, the flat seven, and it happens **40 percent** of the time.

Same current state. Different futures, depending on history. So the property is **false in the strict sense**, and I would rather say that than pretend otherwise.

What saves the model is that I know the **direction** of the error. Look at what those two histories are: sixth down to fifth down to flat seven is a descending line, and it keeps descending. Descending lines carry momentum that a first-order chain cannot see. So the model will systematically under-predict continuation and over-predict turning around.

I did also fit a second-order chain. In sample it looks much better, BIC improves by about 19,000. On held-out data it is **worse**, mean log-likelihood drops from minus 3.36 to minus 3.81. There is not enough corpus to estimate 428 squared rows, so first order is not just a simplification, it is the order the data can actually support.

> IF ASKED
>
> **Then why proceed at all?** Because a first-order chain is the model the data supports, and because the quantities I use it for are aggregate, stationary distribution, hitting times, one strong cell, which are far more robust to this than step-by-step prediction would be.
>
> **Did you test against a null?** Yes, two. Shuffling everything within a tune, and a stricter null that holds the beat sequence fixed and permutes only pitch and chord within each beat class. Observed KL is 2.55 bits; the plain null gives 0.99 and the beat-preserving null gives 1.64. So metre accounts for about 40 percent of the structure, and the chain still beats the strict null with p below 0.001 over 1,000 permutations.

7 A sample walk 1:05

To see what the operator has learned, the simplest test is to let it walk and look at the result next to the real thing.

Left panel. On the left, every step is one complete state, plotted by its index. The states are ordered by chord quality, so a jump between bands is a chord change, and height within a band is the scale degree. Filled markers are on the beat, hollow markers are offbeat.

The interesting number: over the same 100 events, **Parker visits 52 distinct states and the sampled walk visits 81**. The real line keeps returning to a smaller working vocabulary. The model wanders. That gap is essentially the phrase structure the chain cannot represent.

Right panel. On the right is the same walk with two of the three coordinates thrown away, keeping only the scale degree, so it reads as a melodic contour, with rests below the dotted line. It is easier on the eye, but I want to flag that it is a shadow: two points at the same height can be completely different states, differing in chord or in beat.

8 The same walk, engraved 0:45

And here is the same sampled walk written out as notation, so you can judge it as music rather than as a plot.

On top, the pickup and first four bars of *Anthropology* as Parker played them. Underneath, the opening of the sampled walk, entering on the downbeat of bar two, over the identical chord changes.

My honest reading is that it holds up **bar to bar**. The intervals are plausible, the notes land in sensible places against the chords, and it would not immediately give itself away. What it does not have is any sense of a phrase: it never sets up an idea and answers it, and over a full chorus that absence is obvious.

One caveat: the model carries no durations, so the rhythm here is imposed and the rest lengths are randomized. What is genuinely from the model is the sequence of pitches against chords and their beat positions.

9 Stationary distribution

1:20

Now the first of the four results. Let the walk run for a long time and count the fraction of time it spends in each state. That fraction settles down to a fixed number per state, and the vector of those numbers is **pi**: one number per state, summing to one.

Algebraically, pi is the distribution that **one more step leaves unchanged**. It is the left eigenvector of P-hat at eigenvalue 1, and by Perron and Frobenius it exists and is unique because the chain is irreducible and aperiodic.

Point at the three busiest states. The three busiest states are all dominant seventh, all offbeat, and none of them is a chord tone. The most-occupied state in Parker's language is the fifth of a dominant chord on the "and of four".

Point at the scatter. And here is why this is worth showing. Every dot is a state: pi on one axis, and how often that state actually occurs in the corpus on the other. They sit on the diagonal.

That is not circular, and it is the part I want to be clear about. **Pi is computed only from the transitions.** It never sees how common each state is. It comes out of the eigenvector of a matrix of conditional probabilities. And yet it reproduces the corpus frequencies almost exactly.

So the ergodic theorem holds here, empirically: the long-run behaviour of one walk equals the average over the ensemble. It is a check that the estimation is sound before I ask the operator anything harder.

> IF ASKED
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> **How do you know it is irreducible and aperiodic?** Smoothing makes every entry strictly positive, so the chain is trivially both. Without smoothing it is still irreducible on the observed states, but the guarantee is what smoothing buys.

10 A thousand agents

0:50

The previous slide computed pi algebraically. This one gets it the other way, by simulation, which is the version that fits the course.

Drop **1,000 agents** on random starting states. Each one takes **100 steps**, choosing where to go by the odds in the row it is currently in. Nothing else. That is 100,000 events of newly generated Parker.

Then count where they collectively spent their time and compare with pi. *Point at the scatter.* Correlation **0.998**.

So the two routes to the same object agree: solving the eigenvector problem, and just letting a thousand agents run. The operator does what the theory says it should.

11 The walk on a graph 0:40

The matrix and a graph are the same object. 428 nodes, one per state, and a directed edge from i to j weighted by the probability of that move.

428 nodes is not something you can look at, so this is the subgraph on the **eight states with the largest pi**, the top of the list from two slides ago. Between them they account for 9.7 percent of the walk's time.

Two things about those eight, and neither was designed in. **Every one of them is a dominant seventh state**, and **seven of the eight sit offbeat**. The busiest places in Parker's language are unstable notes on weak positions. That is not where a naive model would put the centre of gravity, and it is a first hint of the result on slide 13.

12 PageRank 0:50

PageRank is pi with a teleport. It is the same walk with one change: 85 percent of the time follow an edge as usual, and the other 15 percent of the time jump to a state chosen at random. The stationary distribution of that modified walk is the PageRank score.

The teleport is what makes it well behaved. It guarantees the chain is irreducible and aperiodic, so there is a unique answer, and it stops the ranking being dominated by any part of the graph the walk could get stuck in.

Point at the left panel. The top of the PageRank list is the fifth of a dominant chord on the "and of four". Nine of the ten are dominant sevenths and eight sit offbeat.

Point at the right panel. Next to it, **betweenness**, which asks a different question of the same graph: not where the walk spends time, but which states sit on the likely routes between other states. It gives almost the opposite answer. **Eight of its top ten are rests**, and nine of them sit on the beat.

So offbeat notes are where the line **lives**, and rests on the beat are where it **passes through**. That second one is worth a sentence: a rest is where one phrase ends and the next begins, so silence is the junction between parts of the vocabulary that otherwise do not connect.

> IF ASKED
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> **Isn't offbeat just more common anyway?** Yes, largely, and do not oversell this slide. 55 percent of events are offbeat and 55 percent are on dominant chords, and pi puts almost exactly that much mass in each, so on aggregate the walk is offbeat about as often as the corpus is. Eight of ten in the top of the PageRank list is above the 5.5 you would expect by chance, but only mildly. **The betweenness side is the real signal**: one of ten offbeat is far below chance. And the strong offbeat result in the project is slide 13, where all eight fastest-resolving states are offbeat, which cannot be a base-rate effect because it ranks by hitting time rather than by frequency.
>
> **Why does betweenness weight edges by minus log P?** Because shortest-path algorithms add weights while probabilities along a path multiply. The log turns the product into a sum, and the minus sign makes a high probability a low cost, so the cheapest path is the most likely one.
>
> **How does PageRank compare to the stationary distribution?** Very close, correlation 0.989. With damping 0.85 on a well-connected graph PageRank is a mild perturbation of pi, so it is a robustness check more than a new result.
>
> **Did you run walks to get this?** No. It is power iteration on the damped operator, run to a tolerance of 1e-12. The interpretation is the average over infinitely many walks, but it is computed exactly rather than sampled. The only sampling in the talk is slide 10.

13 Hitting times

This is the result I would defend first, because it turns a metaphor into a measured quantity.

Musicians say some notes are **unstable** and want to resolve. That is a statement about time: an unstable note is one you do not stay on. So define resolution, then measure how long it takes.

I mark a set of states as resolved: **a chord tone landing on one of the four beats**. That is 73 of the 428. Then, from every other state, I ask: starting here, how many events pass before the line first lands in that set? That is the expected hitting time, and it comes from solving one linear system, **h equals (I minus Q) inverse times the all-ones vector**, where Q is the operator restricted to the unresolved states.

Point at the numbers. A typical state resolves in **4.1 events**. The slowest takes 5.1. But the fastest takes **1.84**, which is less than two events away.

Point at the list. Now look at which states those are. The major seventh over a minor chord. The flat second over a dominant. The flat second over a major seventh.

Every one of them is a **semitone away from a chord tone**. These are precisely the notes theory calls **approach tones**, and it is the standard bebop device: play the note next door, then step into the chord tone.

The model was never told which notes are chord tones, never told what a semitone is, never told anything clashes. It counted. And the notes it ranks as most urgent are exactly the ones a teacher would name.

One more thing, and it is the third piece of the research question. **All eight of the fastest-resolving states sit offbeat.** So the pull toward resolution is not just toward certain notes, it points at the downbeat. Tension is placed off the beat and released on it, which is the asymmetry I set out to look for.

> IF ASKED
>
> **Is the result an artefact of how you defined R?** Fair challenge, and I tested it. If R is narrowed to chord tones on major seventh chords only, the median hitting time jumps from 4.1 to about 35 events, but all six fastest states are already on major seventh chords, so that version measures how long until the next major chord, which is harmonic rhythm, not gravity. The current definition is the one that isolates the effect I am after.
>
> **Why an expectation and not a distribution?** The full distribution is available from the same system; the mean is what makes the ranking legible on a slide.

14 Absorption probabilities

The previous slide asked how long resolution takes. This one asks **where it lands**. Take an unresolved note, follow the line forward until it first hits a chord tone on a beat, and record which one.

Walk the four steps, do not read them. Split the 428 states into the 73 resolved ones, call that R, and the other 355, call that T. Cut two blocks out of the operator: **Q**, the moves from unresolved to unresolved, and **P T to R**, the moves that land in one step. Solve one linear system. Each row of the answer is a distribution over the 73 resolved states, and it sums to 1 because the line always lands eventually.

The reason this is free is worth thirty seconds, and it is no longer on the slide, so it has to be said. **Inverting I minus Q gives I plus Q plus Q squared and so on**, and entry i, k of that counts how many times the line is expected to visit unresolved state k before it lands. Sum a row and you get slide 13's waiting time. Weight it instead by the one-step chance of landing, and you get the destination. Same matrix, two questions, one inversion.

Point at the bars. Say clearly that the width of a segment is its probability. Four of the six fastest-resolving states resolve **by a semitone onto the nearest chord tone**, between 49 and 66 percent of the time. The fifth, the 2nd over a major chord, falls a whole tone to the root, which is the other standard resolution. Nothing in the model knows what a semitone is or which notes belong to which chord.

Point at the bottom bar. This one is worth stopping on. The minor third over a major chord resolving up to the major third, 61 percent. That is the same D-flat to D I showed on slide 3, in the *Anthropology* excerpt, one beat position over. **The example I opened the talk with turns out to be the model's own strongest answer.**

One caveat I would rather state myself: the **pitch** here is voice leading, but the **timing** is metre. Every target sits on the next strong beat, largely because the beat cannot go anywhere else.

> IF ASKED
>
> **Where does the equation come from?** First-step analysis. From i, either you land on r immediately, probability P T-to-R at i r, or you step to some other unresolved k, probability Q at i k, and face the same question from k. So B equals P-T-to-R plus Q B. Move the Q B across and you get (I minus Q) B equals P-T-to-R. The inverse is just isolating B.
>
> **What is I?** The identity matrix, same size as Q, ones down the diagonal and zeros elsewhere. It is the matrix version of the number 1: multiplying by it changes nothing. It appears because the equation is B equals P-T-to-R plus Q B, and to collect the B terms you need B minus Q B. You cannot factor that as one minus Q times B, since 1 is a number and Q is a matrix, so you write I B minus Q B and factor out to (I minus Q) B. In the series I plus Q plus Q squared and so on, the I term is the zero-steps case: the visit you are already making.
>
> **Why is I minus Q invertible?** Because Q's spectral radius is below 1, here 0.74. Every unresolved state has a positive chance of escaping into R, which the alpha = 0.01 smoothing guarantees. Without smoothing a zero row could break it.
>
> **Isn't R already the resolution set? Why call the block P T to R?** Good catch, and deliberate. The textbook writes that block as R, which would collide with R the set of 73 states. Two different objects, one letter, so I renamed the matrix.
>
> **What are the unlabelled segments?** Only the widest segment in each row can hold a name at that size. The second on row one is the flat 7, 16 percent, so that state resolves by semitone up to the root 63 percent of the time and by semitone down to the flat 7 another 16. On row two the second is the flat 7 of the dominant, 24 percent.

15 Voice-leading recovery 1:30

One more question of the operator, and it needs a distinction I should make explicitly.

A row of P-hat can be almost certain about a state Parker hardly ever plays. A rule can be near-deterministic and still be irrelevant, because the situation never comes up. So to ask **which transitions actually carry the music**, one number is not enough.

Point at the formula. Multiply the two things I already have. **Pi at i**, from slide 9, is the share of its time the walk spends at that state. **P-hat at i j**, from slide 5, is the chance the next event is j *given* that it is at i. Their product is the share of **all** the moves in the corpus that are this exact move.

So the two numbers answer different questions. **Conditional** asks: standing here, where next? **Joint** asks: how much of the music is this? Ranking by joint puts what Parker plays **most** at the top, rather than what is most rule-like.

Point at the chart. And here is the ranking. Three things about those twenty, none of them designed in. **Every single one stays on the same chord quality**, so the busiest moves happen inside a chord rather than across a change. **Fourteen move by a step or less**, two semitones at most, and **thirteen descend**. The other six are leaps, and every one of those runs **from one chord tone to another**. So there is nothing in the top twenty that is not either a step or an arpeggiation.

That is a description of a bebop line: mostly stepwise, mostly downward, chromatic, resolving inside the harmony. It came out of ranking a matrix by traffic.

> IF ASKED
>
> **Why is the joint so small, 0.37 percent at the top?** Because it is a share of all 21,540 transitions across 428 states. Uniform would be about 0.0005 percent, so the top cell is roughly 700 times uniform. Small in absolute terms, enormous relative to chance.
>
> **What about the textbook V7 to I resolution?** It is in the operator and it is sharp: from the flat 7 of a dominant on the "and of four", the next note is the major 3rd of the tonic on the downbeat **10.71 percent** of the time, with the cells around it all under 0.1 percent. It is not in this top 20 because it needs a particular chord change under a particular note, so it is rare in absolute terms even though it is near-certain when it arises. That contrast is the joint-versus-conditional point in one example.
>
> **Which neighbouring cells, exactly?** The same move starting from beat 4 instead of the "and of four" is 0.05 percent; from the "and of three", 0.09; landing on beat 2 instead of beat 1, 0.01.
>
> **Is 10.71 percent large?** Against 428 possible destinations, uniform would be 0.23 percent, so about 45 times uniform, and the neighbours under identical conditions are 100 to 1,000 times smaller. Smoothing cannot have created it either, since smoothing pulls toward uniform and can only shrink a peak.

16 Questions I didn't have time to address 0:30

Before I sum up, one slide of honesty about scope. An operator like this answers far more questions than fit in twenty minutes, so here are the ones I could have put to it and did not.

Do not read the list. Leave it up and give the shape of it in two sentences.

Roughly they fall into two kinds. Questions about **time**: how often things come back round, whether resolution is reliable or erratic, how long until the walk has played everything it knows. And questions about the **shape of the whole language**: how predictable it is, whether it splits into neighbourhoods, whether it still makes sense backwards, what a line that never lands looks like.

One of them, how often a state comes back round, I did compute after the fact, and there is a backup slide for it. The question missing from this list is where the line resolves to, because that is slide 14.

> IF ASKED
>
> **Pick any question on the slide.** "How long until Parker plays this state again" has a backup slide. For the rest, the honest answer is the method rather than the number, and each is worth naming: the backwards question is the time-reversed chain, the never-resolves question is the quasi-stationary distribution, the surprise question is the entropy rate, and the neighbourhoods question is clustering on the slow eigenvectors.
>
> **Why not just do them?** Each is cheap given what is already computed. The constraint was the twenty minutes, not the arithmetic.

17 Take-aways 1:15

So, back to the question I opened with. Four things came out, and four did not.

Voice leading came out as the busiest traffic. Rank every transition by joint probability and the top twenty in the whole operator are all moves inside a single chord: fourteen by a step or less, thirteen descending, every one of the six leaps chord tone to chord tone. That is a description of a bebop line, and it came out of ranking a matrix by traffic.

Harmonic gravity came out as a waiting time. A typical state reaches resolution in 4.1 events, the most unstable ones in under 2, and the notes that resolve fastest are the ones theory calls approach tones. The chain found them without being told which notes clash.

And gravity turned out to have a destination, not just a delay. The same linear system says which note the line lands on, and four of the six fastest-resolving states land by a semitone on the nearest chord tone, between 49 and 66 percent of the time. Again nothing told the model what a semitone is.

And the on-beat, offbeat asymmetry came out everywhere. All eight of the fastest-resolving states sit offbeat, and every one of them resolves onto a strong beat. Tension is placed off the beat and released on it, which was the third piece of the research question.

Move to the right column. Now what did not work, because I think this half is the more useful one.

A Markov chain may not be the right tool. I showed on slide 6 that the same state behaves differently depending on how the line arrived, and second order is not affordable on this much data. The assumption is convenient rather than true.

Much of the structure is metre, not music. The matrix is 97 percent empty largely because the beat cannot jump. Against a null that holds the beat sequence fixed, roughly 40 percent of the apparent structure disappears. Some of what looks like syntax is just bar lines, and I would rather say that myself than have it asked.

The generated lines are not Parker. Right interval sizes, right rest frequency, no phrases. The engraved staff is the honest version: convincing bar to bar, falling apart across a chorus.

And most of a note is missing. No register, no duration, no dynamics. The state is a pitch class against a chord on a beat, and nothing else. What is remarkable is how much came out of so little, but it is worth being clear about how little that is.

18 Further work 0:25

Very briefly, the obvious next steps, and I will leave this on screen rather than read it.

The main one is **splitting the operator** and comparing the halves, by whether the chord changes, by direction, or by phrase position. The bar a split has to clear is that it conditions on something not already in the state and not explained by metre. Direction already looks promising: descending lines land a chord tone on a strong beat 67 percent of the time against 54 for ascending.

Beyond that, adding coordinates the state currently throws away, and a different encoding based on interval content rather than degrees.

> IF ASKED
>
> **Why did you not do the within-chord versus across-chord split?** Answer honestly: I built it and set it aside. Only about 16 percent of transitions cross a chord change, which is arithmetic rather than a finding, since chords change every 6.2 events on average. It is confounded with the downbeat. And 76 percent of the chord-change flag is already recoverable from the state itself, so the split was largely re-cutting information the operator already had.

19 Thank you 0:15

That is the project. Thank you, and I am happy to take questions.

Three backup slides follow. Do not advance into them unless a question takes you there: distribution evolution, the spectrum, coming back round.

B1 Evolving the distribution

on demand

Go here if asked how fast the chain converges, or how you know pi is really the limit.

Slide 9 found pi algebraically, as an eigenvector. This finds it by iteration instead. Start with all the probability on one state, meaning complete certainty about the current note, then multiply by P repeatedly and watch the certainty dissolve.

Starting from the busiest state, the total variation distance to pi is already below 0.5 after **two steps**, below 0.01 after **21**, and by step 40 the largest disagreement with pi anywhere is under nine parts in a million.

The decay tracks the second eigenvalue modulus raised to t almost exactly, so this is the mixing time measured rather than asserted.

B2 The spectrum

on demand

Go here if asked about eigenvalues, mixing time, or periodicity. Worth volunteering if the questions are thin.

Every step the walk forgets a little of where it started, at a fixed rate set by the second eigenvalue. Here its modulus is 0.814, so whatever you still know shrinks to about 81 percent of itself each step, which gives a mixing time of about **5.4 steps**. After roughly five events the chain has largely forgotten where it began.

But the interesting part is that the second eigenvalue is **complex**, 0.596 plus 0.555 i. A real eigenvalue would mean the walk simply fades toward pi. A complex one means it also **circles**: it returns to similar territory at a regular interval, and the interval is two pi divided by the argument.

That comes out at **8.4 events**. One bar of eighth notes is exactly 8 events.

So the metre is sitting in the spectrum of the operator, and nothing put it there. The matrix was built by tallying which state follows which. The bar line came out as the phase of an eigenvalue.

It also tells you the chain is **not reversible**, since a reversible chain has a real spectrum. Which is right: music runs forward.

B3 Coming back round

on demand

Go here if asked how often a state recurs, or about return times as opposed to hitting times. Slide 14 covered where the line resolves to; this is the other half.

Hitting time looks forward to a set the walk has not reached yet. **Recurrence time** asks the opposite: standing on a state now, how many events before the line plays it again. For an irreducible chain that is Kac's formula, one over pi, so it needs nothing beyond the stationary distribution.

The busiest state, the fifth of a dominant on the pickup, comes back every **71 events**, which is about nine bars of eighth notes, so roughly once a chorus. The median state waits **809**. So there is a small core vocabulary that recurs constantly and a long tail that is genuinely occasional.

Point at the spike on the right, before anyone else does. That spike is not a musical fact. Those are states with almost no data, whose pi is mostly smoothing, so their recurrence times are telling you about alpha rather than about Parker.

B4 Questions with no slide

on demand

Answers to keep in your pocket. There is no slide for these.

> IF ASKED
>
> **Why notated scores and not MIDI or audio?** Deliberate. I want the beat position a musician intends, not the microtiming of a performance. Parker plays behind and ahead of the beat constantly, and against a performed reference almost every note would be offbeat. The notation is the syntactic layer, which is the object of the study.
>
> **Why five chord qualities, and why keep diminished separate from half-diminished?** Because their chord-tone sets differ on the seventh degree, so merging them would put a chord tone and a non-chord tone in the same state.
>
> **Fifty solos, one player. How general is this?** It is not, and I would not claim it is. This is a model of Charlie Parker, not of jazz. The method transfers directly to another player's corpus, and comparing operators across players is the natural extension.
>
> **Is the whole thing reproducible?** Yes. One command runs the entire pipeline end to end in about a minute, with a fixed random seed throughout.
>
> **How does this connect to the course?** It is the random-walk-on-a-graph exercise, applied to a real corpus instead of a synthetic graph: transition operator, stationary distribution via Perron-Frobenius, spectral gap and mixing time, hitting times by first-step analysis, PageRank as a damped walk, and Monte Carlo simulation of the ensemble.